

Why Do Onions Make You CRY? ☐☐

Onion = Underground Bulb with a Sulfur
SHIELD ☐☐

**Inside the cell: 2 chemicals locked in
SEPARATE rooms □**

CUT! → Cell walls break → chemicals MIX



Step 1: Enzyme makes SULFENIC ACID
(unstable!) ☐☐

**Step 2: LFS enzyme → tear gas:
syn-Propanethial-S-oxide □**

**Gas + eye water → mild SULFURIC ACID
→ stings nerves ☐**

**Brain → ALERT! → Lacrimal glands make
tears to wash it out ☐☐**

**Onion tears = REFLEX tears (not
emotional!) = wiper**

Tip 1: Chill it ☐

Tip 2: Use a SHARP knife ☐

More tips: □ water

□ flame

□ QUICK BRAIN CHECK! □□ 1. Which 2 things stay separate in the cell? 2. Name the tear gas + the enzyme that makes it 3. What really stings — the gas or what it forms?

□ ANSWERS: 1. Alliinase enzyme + sulfoxides (isoalliin) 2. syn-Propanethial-S-oxide, made by LFS 3. Gas + eye water → mild sulfuric acid stings nerves

RECAP done! □ Tearless onion exists! □
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